



Batterjee Medical College
For Science and Technology

Parathyroid Disorders

Internal Medicine Department

Knowledge

- Relate pathophysiological basics and etiology of parathyroid disorders
- Describe epidemiology, manifestations, complications and management of hypoparathyroidism
- Describe causes, manifestations of hypocalcemia

Skills

- Interpret clinical manifestations and investigations of parathyroid disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for hypoparathyroidism
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

- Hypoparathyroidism (hereditary and acquired causes)
- Clinical picture of hypoparathyroidism (acute tetany, chronic latent tetany, ectodermal and CNS manifestations)
- Investigations and treatment of hypoparathyroidism
- Differential diagnosis of hypocalcemia and tetany (hypocalcemia, alkalosis, magnesium tetany)
- Pseudohypoparathyroidism and pseudo-pseudohypoparathyroidism (clinical and biochemical features)



Hypocalcemia/ Tetany

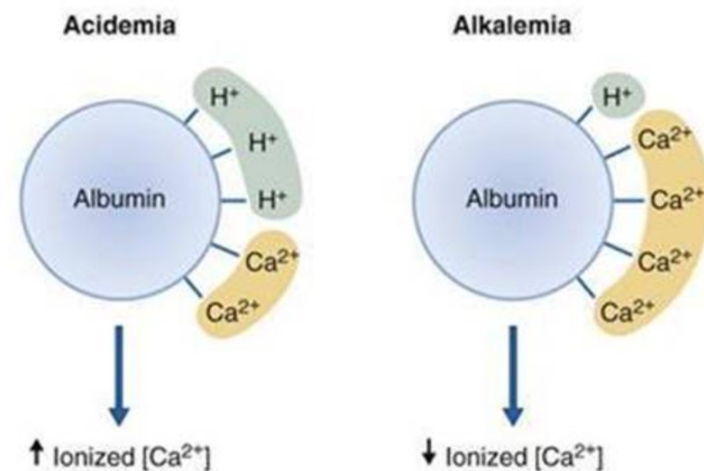
- Definition of tetany: State of ↑ neuro-muscular excitability
- Causes of tetany (intermittent muscular spasms)

1.Hypocalcaemia: (quantitative ↓ of serum Ca).

- A. Hypoparathyroidism.
- B. ↓ Ca intake: as in prolonged starvation, anorexia, dysphagia.
- C. Defective absorption of Ca : e.g.
 - Fatty diarrhea (steatorrhea) → fat combines with Ca forming Ca soaps.
 - Vit. D deficiency: as in rickets & osteomalacia.
- D. Severe acute hyperphosphataemia : due to acute renal failure & tumour lysis.
- E. Precipitation of Ca in tissue: Acute pancreatitis.

2. Alkalosis : (qualitative change i. e. total Ca is kept constant but the ionisable fraction is $\downarrow$ & P is normal) e.g. in :

- Excessive ingestion of alkalies.
- Frequent vomiting.
- Hypokalaemic alkalosis
e.g. Conn's syndrome.
- Hyperventilation (hysterical).



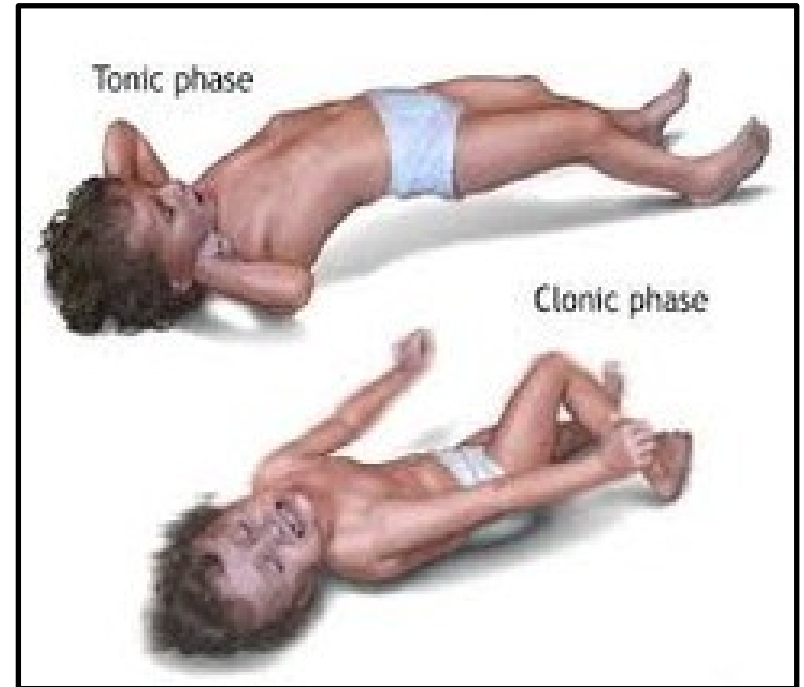
3. magnesium tetany : Mg $\downarrow$ occurs in :

- Prolonged. IV infusion without Mg supplementation.
- Severe diarrhea.
- Malabsorption syndrome.

Clinical picture

Overt (manifest) tetany as 'serum Ca <7 mg%'

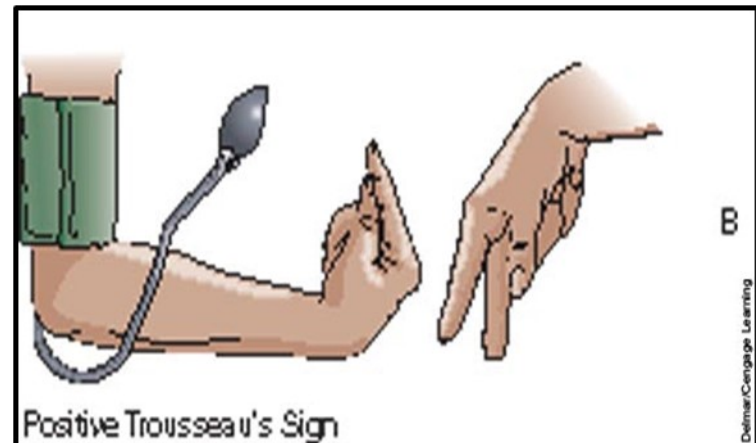
- A. a. Parasthesia of the extremities & around the mouth.
- B. b. Muscle spasm of the hands & feet (carpo-pedal spasm). These are painful spasmodic attacks affecting the carpal pedal muscles ;
 - o In the hands → spasmodic adduction of the thumb, flexion of the wrist & metacarpo-phalangeal joints with extension of the fingers.
 - o The feet show → planter flexion of the toes with inversion & the patient may fall on his face.
- C. Spasm of other muscles :
 - o Laryngospasm → respiratory obstruction → sudden dyspnea in children (laryngismus stridulous) & may be death .
 - o Facial & masseter muscles → risus sardonicus.
 - o Muscles of the back → opisthotonus.
- D. Tonic & clonic spasms : may end in generalized convulsions



Latent tetany 'serum Ca : 7 - 8.5 mg%'

A- Can be provoked by :

- Chvostek's Sign → tapping over the facial nerve in front of the ear → twitches of facial muscles.
- Trousseau's sign: → obstruction of the radial pulse by sphygmomanometer for 4 minutes → carpal spasm (main d'accoucheur).
- Erb's Sign : → < 4 milliamperes produce muscles contraction (normally, at least 8 milliamperes are required).

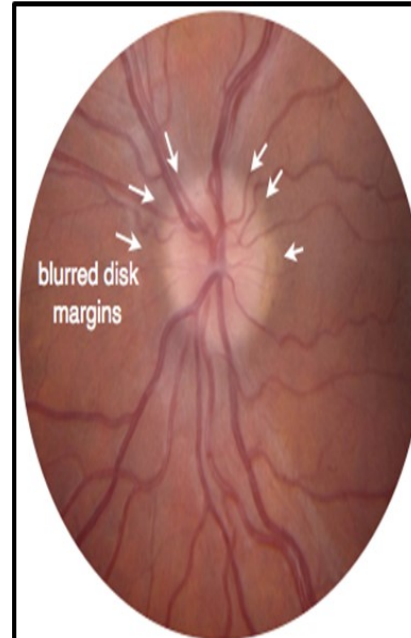


B. manifestations:

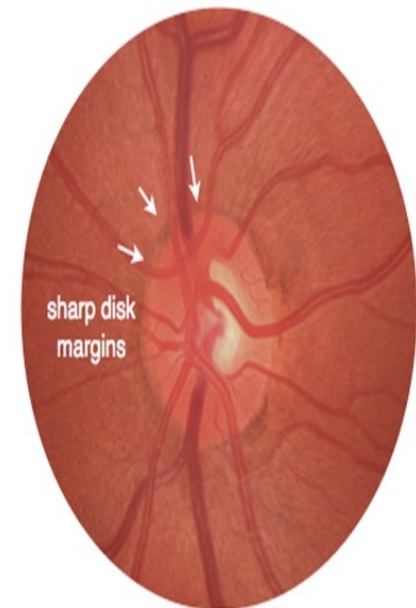
- Skin (only in hypocalcemia)
→ dry, sparse hair, brittle nails & hypoplastic teeth.
- CNS → irritability, anxiety, psychosis & epilepsy.
- CVS → hypotension, heart failure (refractory), prolonged Q-T interval, arrhythmias
- Eyes :
 - Cataract and papilloedema
- Articular → Periarticular calcifications (chondrocalcinosis)

Characteristics of Brittle Nails

- Feeling of dryness
- Nails break, split, or chip easily
- Nails are sunken in appearance
- Nails curl around the finger tip
- Color changes
- Ridges on nail



Papilledema



Normal Optic Disk

Investigation : (for the cause)

- In hypoparathyroidism : $\downarrow$ PTH , $\downarrow$ serum Ca , $\uparrow$ P.
- In hypomagnesaemia : Refractory hypokalemia.
- In alkalemia : $\uparrow$ PH , $\downarrow$ ionized Ca.

Treatment :

a. Acute attack : IV **Ca gluconate** 10 ml 10% very slowly.

b. Treatment of the cause :

- Hypocalcemia : oral Ca , vitamin D
- Hypomagnesemia : oral Mg.
- Alkalosis : Treatment of the cause , acidifying drug



Hypoparathyroidism

1. Hereditary : which may be

- Isolated or combined with other endocrine gland deficiencies e. g. ovaries, adrenals, ... etc. it is due to auto-antibodies " polyglandular autoimmune deficiency syndrome "
- In association with absent thymus "Di George's syndrome ".

2. Acquired :

- Accidental surgical removal (during thyroidectomy).
- Following removal of a parathyroid adenoma (due to suppression of the remaining 3 glands by the high Ca levels) "temporary".
- intraglandular hge in the new born (Tetania neonatorum).
- After I131 ttt of the thyroid, hemochromatosis.

CP:

1.Acute hypoparathyroidism :

Usually presents as manifest tetany

2.Chronic Hypoparathyroidism :

Usually characterized by latent tetany + ectodermal + CNS changes.

NB:

- Soft tissue deposition of Ca is potentiated by hyperphosphatemia present in PTH ↓ 'but not in Vit D ↓ where Ca ↓ is associated with normal or ↓ P level'.

Investigations :

- ↓ serum ca.
- ↑ serum P.
- ↓ serum PTH.

Treatment :

A. Acute hypoparathyroidism (tetany) :

Ca gluconate 20 ml 10 % solution IV (can be ↑ to 100 ml) / 4 Hrs.

B. Maintenance therapy:

1. Ca gluconate → 1.5- 3 gm t.d.s. +
2. Vit. D (calciferol): 'or better, dihydrotachysterol, AT. 10'
50,000 IU. / day (1,25 mg) " 1 mg = 40,000 I.U. ". or
3. 1 α Vit.D3 (calcitriol) : 0.2 - 1 ug / d.
4. Thiazides → ↓ urinary Ca excretion.
5. Treatment of the cause of alkalosis (if present).

Differential diagnosis

I. From other causes of hypocalcaemia & tetany

II. Other types of Hypoparathyroidism

1.Pseudohypoparathyroidism :

- Autosomal dominant inheritance characterized by peripheral resistance to the action of PTH (its level may be normal or $\uparrow$).
- All physical & biochemical features of Hypoparathyroidism.
- Subnormal intelligence & basal ganglia calcification are more frequent.
- Additional characteristic features :
 - Round face - Short stature - obesity - Short metacarpal bones

2.Pseudo—pseudohypoparathyroidism:

- This is an incomplete form of pseudohypoparathyroidism which may be present in siblings of affected patients.
- Only the phenotypic features are present without hypocalcaemia or hyperphosphataemia.



Condition		AHO	PTH levels	Calcium	Phosphates	Urinary Calcium*	Molecular Defect
Hypoparathyroidism		NO	Low	Low	High	High	Variable
Pseudohypoparathyroidism	Type 1A	YES	High	Low	High	Low	GNAS defect on the maternal allele
	Type 1B	Mild	High	Low	High	Low	Imprinting defect on maternal GNAS allele
	Type 2	NO	High	Low	High	Low	NIL
Pseudopseudohypoparathyroidism		YES	Normal	Normal	Normal	Normal	GNAS defect on paternal allele

Laboratory differential of common causes of Hypocalcemia

Serum levels	Hypoparathyroidism	Vitamin D deficiency	Pseudohypoparathyroidism
Serum PTH	↓	↑	Normal or ↑
Serum Calcium	↓	↓	↓
Serum Phosphate	↑	↓	↑

Hypocalcemia

Definition

Mild hypocalcemia: total serum Ca^{2+} < 8.5 mg/dL
or ionized Ca^{2+} < 4.65 mg/dL
Severe hypocalcemia: total Ca^{2+} ≤ 7.5 mg/dL
or ionized Ca^{2+} < 3.6 mg/dL
Factitious: decreased total Ca^{2+} with
normal ionized Ca^{2+} (asymptomatic)

Etiology

With low PTH: hypoparathyroidism
With high PTH: vitamin D deficiency, CKD,
acute pancreatitis, hyperphosphatemia
Other: medications (e.g., loop diuretics),
multiple blood transfusions, hypomagnesemia

Diagnosis

Serum total and ionized Ca^{2+} , electrolytes and
kidney function, ABG/VBG, serum intact PTH
ECG (e.g., prolonged QT interval)

Treatment

Calcium supplementation
Treatment of the underlying condition

Neurological

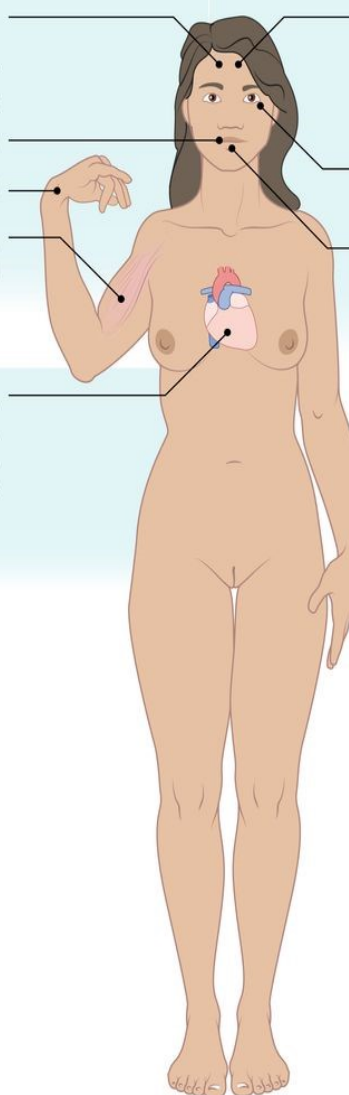
Seizure
Tetany (increased neuro-
muscular excitability):
Paresthesias (e.g., perioral)
Spasms (e.g., carpopedal)
Muscle cramps,
stiffness, myalgia

Cardiovascular

Arrhythmias (palpitations,
irregular pulse, syncope),
hypotension,
congestive heart failure

Chronic manifestations

Psychiatric:
e.g., anxiety, depression
Neurological:
pseudotumor cerebri
Ophthalmologic:
papilledema, cataracts
Dental changes:
altered morphology,
enamel hypoplasia



Note

Symptoms of tetany (e.g., spasms, cramps and paresthesia) are the most characteristic features of hypocalcemia.

A 68-year-old man comes to the physician because of fatigue and muscle cramps for the past 4 weeks. He has also noticed several episodes of tingling in both hands. He has not had fever or nausea. He has had a chronic cough for 10 years. He has chronic bronchitis, hypertension, and osteoarthritis of both knees. His father died from lung cancer. Current medications include salbutamol, ibuprofen, and ramipril. He has smoked 1 pack of cigarettes daily for 45 years. He is 175 cm (5 ft 9 in) tall and weighs 68 kg (163 lb); BMI is 22 kg/m². His temperature is 36.7°C (98°F), pulse is 60/min, and blood pressure is 115/76 mm Hg. While measuring the patient's blood pressure, the physician observes carpopedal spasm. Cardiopulmonary examination shows no abnormalities. His hematocrit is 41%, leukocyte count is 5800/mm³, and platelet count is 195,000/mm³. Serum alkaline phosphatase activity is 55 U/L. An ECG shows sinus rhythm with a prolonged QT interval.

Which of the following is the most likely underlying cause of this patient's symptoms?

- a) Multiple endocrine neoplasia
- b) Ectopic hormone production
- c) Medication side effect
- d) Destruction of parathyroid glands
- e) Albright hereditary osteodystrophy
- f) Vitamin D deficiency

A 41-year-old man comes to the emergency department with a 2-day history of nausea, abdominal discomfort, and diarrhea. His abdominal discomfort worsens shortly after meals. He has also had progressive perioral numbness and upper extremity muscle cramping for the past 24 hours. He underwent a total thyroidectomy to treat a Hurthle cell lesion 4 days ago. His mother died of lung cancer at age 68. He has smoked one pack of cigarettes daily for 22 years. His only current medication is a multivitamin supplement. He appears tired. Temperature is 36.0°C (96.8°F), pulse is 72/min, respirations are 20/min, and blood pressure is 130/70 mm Hg. Inflating the arm cuff while taking the patient's blood pressure triggers a carpal spasm. Physical examination shows a well-healing surgical wound on the neck. The abdomen is mildly tender to palpation. The remainder of the examination shows no abnormalities. Serum studies show:

Na ⁺	138 mEq/L	K ⁺	4.2 mEq/L	Cl ⁻	102 mEq/L
HCO ₃ ⁻	25 mEq/L	Mg ²⁺	1.7 mg/dL	Phosphorus	4.3mg/dL
25-hydroxyvitamin D	20 ng/mL (N = 20–100)				
Parathyroid hormone	2 pg/mL				

Which of the following is the most appropriate initial step in pharmacotherapy?

- a) Oral calcium citrate
- b) Intramuscular calcitonin
- c) Intravenous magnesium
- d) Intravenous calcium gluconate
- e) Oral vitamin D
- f) Subcutaneous parathyroid hormone

A 63-year-old man comes to the physician because of fatigue and muscle cramps for 6 weeks. He also noticed several episodes of tingling around the mouth and in the fingers and toes. He has osteoarthritis of his knees and hypertension. Current medications include ibuprofen and ramipril. He has smoked one pack of cigarettes daily for 35 years. Tapping over the facial nerve area in front of the ear elicits twitching of the facial muscles on the same side of the face. His serum alkaline phosphatase activity is normal, serum phosphorus is 5.2 mg/dL, and serum creatinine is 1.1 mg/dL. An ECG shows sinus rhythm with a prolonged QT interval. Which of the following is the most likely underlying cause of this patient's symptoms?

- a) Vitamin D deficiency
- b) Ectopic hormone production
- c) Destruction of parathyroid glands
- d) 1ry hyperparathyroidism

Summary

I just need
the main ideas



References

1. AMBOSS
2. Step-up to Medicine, 6th edition (ISBN-13: 978-1975192730)
3. Davidson's Principles & Practice of Medicine, 25th edition (ISBN-13: 978-0443287626)
4. Harrison's Principles of Internal Medicine, 22nd edition (ISBN-13: 978-1265979317)
5. Current Medical Diagnosis & Treatment, 65th edition (ISBN-13: 978-1260469868)
6. Kumar and Clark's Clinical Medicine, 11th edition (ISBN-13: 978-0443128288)

Thank You